MISCELLANEOUS

123. A city has a population of 3,00,000 out of which 1,80,000 are males. 50% of the population is literate. If 70% of the males are literate, the number of literate females is:

(a) 24,000 (b) 30,000
(c) 54,000 (d) 60,000

Ans. (a) : Total population = 300000
Total number of males = 180000
Total Literates = 50% of total population = 150000
Number of literate males = 70% of males = 126000
∴ Number of literate females = 150000 – 126000 = 24000

124. In a survey, it was found that 80% of those surveyed owned a car while 60% of those surveyed owned a mobile phone. If 55% owned both a car and a mobile phone, what percent of those surveyed owned a car or a mobile phone or both?

(a) 65% (b) 80%
(c) 85% (d) 97.5%

Ans. (c) : Percentage of car owners = 80%
Percentage of mobile phone owners = 60%
Percentage of people having both car and mobile phone = 55%
Percentage of people having only car = 80 – 55 = 25%
Percentage of people having only mobile phone = 60 – 55 = 5%
Percentage of people having a car or a mobile phone or both = 55% + 25% + 5% = 85%

125. Consider the following statements regarding the Armed Forces:

1. **First batch of women pilots was commissioned in Indian Air Force in 1996.**
2. **Officers' Training Academy is located in Nagpur.**
3. **Southern Command of Indian Navy has its Headquarters at Chennai.**
4. **One of the Regional Headquarters of Coast Guard is located at Port Blair.**

Which of these statements are correct?

(a) 1, 2, 3 and 4 (b) 2, 3 and 4
(c) 3 only (d) 4 only

Ans. (*) : First batch of women pilot was commissioned in 1996. Gunjan Saxena was one of the 6 women pilot inducted in 1996. Officer's training Academy (OTA) is situated in Chennai and not in Nagpur. Headquarter of Southern Command of Indian navy is in Cochin. Regional Headquarter of a coast guard is located at Mumbai, Chennai and Port Blair.

126. Consider the following statements regarding power sector in India:

1. **The installed capacity of power generation is around 95000 MW.**
2. **Nuclear plants contribute nearly 15% of total power generation.**
3. **Hydroelectricity plants contribute nearly 40% of total power generation.**
4. **Thermal plants at present account for nearly 80% of total power generation.**

Which of these statements is/are correct?

(a) 1 only (b) 2 and 3
(c) Third (d) Fifth

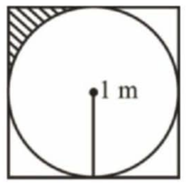
Ans. (a) :
According to the data of 2000, the power generation is 95000 MW. Nuclear plants contribute 2% of total power generation. Hydroelectricity plants contribute 25% of total production and 73% is contributed by thermal power plants.

127. In 1930, a person's age was 8 times that of his son. In 1938, the father's age became ten times that of his son's age in 1930. The ages of the son and father in 1940 were, respectively:

(a) 16 years, 59 years (b) 15 years, 50 years
(c) 14 years, 42 years (d) 13 years, 34 years

Ans. (c) : Let the father's age and his son's age in 1930 be $8x$ and x respectively.
In 1938, father's age = $(8x+8) = 10x$
$2x = 8$
$x = 4$
Age of the son in 1940 = $(x+10) = 4 + 10 = 14$
Age of the father in 1940 = $(8x+10) = 8\times4 + 10 = 42$

128.



A circle of 1 m radius is drawn inside a square as shown in the figure given above. What is the area of the shaded portion in m^2?

(a) $(4-\pi)$
(b) $\left(1-\frac{\pi}{2}\right)$
(c) $\frac{1}{4}-\frac{\pi}{4}$
(d) $\left(1-\frac{\pi}{4}\right)$

Ans. (d) : Since radius of circle = 1 m
∴ side of the square = 1 + 1 = 2m
So area of circle = $\pi r^2 = \pi$
And area of square = $(2)^2 = 4m^2$
Area of shaded region = $\frac{1}{4}$(Area of square – Area of circle)
$= \frac{1}{4}(4-\pi)$
$= \left(1-\frac{\pi}{4}\right)m^2$

129. Match List I with List II and select the correct answer using the code given below the lists:

List I (Important Day)	List II (Date)
A. World Environment Day	1. March 20
B. World Forestry Day	2. June 5
C. World Habitat Day	3. September 16
D. World Ozone Day	4. October 3
	5. December 10

	A	B	C	D
(a)	2	1	4	5
(b)	1	2	4	3
(c)	1	2	3	4
(d)	2	1	4	3

Ans. (d) : The correct match is -
A.World Environment Day - June 5
B. World Forestry Day - March 20
C. World Habitat Day - October 3
D. World Ozone Day - September 16

130. Match List I with List II and select the correct using the codes given below the lists:

List I (Books)	List II (Author)
A. The Struggle in my life	1. Lech Walesa
B. The Struggle & the Triumph	2. Nelson Mandela
C. Friends & Foes	3. Leonid Brezhnev
D. Rebirth	4. Zulfikar Ali Bhutto
	5. Sheikh Mujibur Rehman

	A	B	C	D
(a)	1	2	5	3
(b)	2	1	5	3
(c)	1	2	3	4
(d)	2	1	3	4

Ans. (b) : The Struggle in My Life is written by Nelson Mandela, " The Struggle and the Triump" is written by Lech Walesa, "Friends and Foes" is written by Sheikh Mujibur Rahman and "Rebirth" is written by Leonid Brezhenev, the former president of U.S.S.R

131. Match List I with List II and select the correct answer using the codes given below the lists:

List I (Achievement in genetics)	List II (Scientists)
A. Discovery of transduction and conjugation in bacteria	1. Khurana
B. Establishing the sex-linked inheritance	2. Kornberg
C. Isolation of DNA polymerase from E. Coli	3. Lederberg
D. Establishing the complete genetic code	4. Morgan
	5. Ochoa

	A	B	C	D
(a)	4	3	2	1
(b)	3	4	1	5
(c)	4	3	1	5
(d)	3	4	2	1

Ans. (d) : The correct match is option (d)

A.	Discovery of transduction and conjugation in bacteria	1. Lederberg
B.	Establishing the sex-linked inheritance	2. Morgan
C.	Isolation of DNA polymerase from E. Coli	3. Kornberg
D.	Establishing the complete genetic code	4. Khurana

132. Anand must be a vegetarian because he is a Buddhist, The argument assumed that:

(a) Most Buddhists are vegetarians
(b) All Buddhists are vegetarians
(c) Only Buddhists are vegetarians
(d) Most vegetarians are Buddhists

Ans. (b) : Anand must be a vegetarian because he is a Buddhist, The argument assumed that all Buddhists are vegetarians.

133. Most guitarists are bearded males. If a represents all males, B represents beared males and C represents all male guitarists, then the correct diagram for their relation (shaded portion) is